

Callum T. Donnan

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Research Interests

Galaxy formation & evolution, first stars & galaxies, cosmic reionization, stellar populations, large-scale structure, galaxy-halo connection

Education

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| PhD Astrophysics , University of Edinburgh, Edinburgh, UK | 2021 - 2025 |
| • Thesis: " <i>Exploring the formation of the first galaxies with VISTA and JWST</i> " | |
| • Supervisors: James Dunlop, Ross McLure, Derek McLeod | |
| MPhys Astrophysics , University of St Andrews, St Andrews, UK | 2016 - 2021 |
| • Thesis: " <i>The gas content of galaxies in the cosmic web</i> " | |
| • Supervisor: Rita Tojeiro | |

Research Positions

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| Post Doctoral Research Associate , NOIRLab, Tucson, AZ, USA | 2025 - Present |
| Summer Research Student , University of St Andrews, St Andrews, UK | 2021 |
| Institute of Astronomy Summer Research Programme , University of Cambridge, Cambridge, UK | 2020 |
| Summer Research Student , University of St Andrews, St Andrews, UK | 2019 |

Awards & Grants

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| JWST Cycle 4 PI GO 6954 , ~ \$95k | 2025 |
| MPhys Thesis Prize for best MPhys astrophysics thesis | 2021 |
| Dean's List (awarded for averaging a first class across the academic year.) | 2016 - 2021 |
| Cormack Vacation Scholarship from the Royal Society of Edinburgh | 2019 |
| School of Physics & Astronomy Student Staff Council Vacation Award | 2019 |
| Medal (Astronomy and Astrophysics First & Second Level) | 2016 - 2018 |
| Margaret Stewart Prize (Awarded to the best First Year student in Astronomy and Astrophysics) | 2016 - 2017 |

Telescope Programs

Co-I , Is the quest for Cosmic Dawn beyond all H0pe? , GO 11171, 13 hours, PI: D. McLeod	JWST Cycle 5
PI , Spectroscopic confirmation of the most robust $z=15$ galaxy candidate , GO 6954, 6 hours	JWST Cycle 4
Co-I , The CANDELS-Area Prism Epoch of Reionization Survey (CAPERS) , GO 6368, 194 hours, PI: M. Dickinson	JWST Cycle 3
Co-I , Dead or alive? Unveiling the nature of massive galaxies in the early Universe , GO 5545, 10 hours, PI: L. Barrufet	JWST Cycle 3
Co-I , EXCELS: The Early eXtragalactic Continuum and Emission Line Survey , GO 3543, 72 hours, PI: A. Carnall & F. Cullen	JWST Cycle 2
Co-I , Characterizing true $z > 12$ galaxies and $z \sim 4 - 7$ interlopers in preparation for JWST Cycle 2 , DD 2750, 8 hours, PI: P. Arrabal Haro	JWST DDT
Co-I , The AURORA Survey: First Direct Metallicity Calibrations at High Redshift , GO 1914, 63 hours, PI: A. Shapley & R. Sanders	JWST Cycle 1
Co-I , PRIMER: Public IMaging for Extragalactic Research , GO 1837, 196 hours, PI: J. Dunlop	JWST Cycle 1

Supervision, Teaching & Outreach

Public talk , Space drafts	2026
Supervisor , undergraduate summer research project	2024
Teaching Assistant , Physics 1A, Discovering Astronomy, Observational Astronomy	2021-2024
Royal Observatory open day , JWST presentation to members of the public	2023
High School Outreach , presentations to high school students on astronomy	2018-2023

Invited Talks

Miracles of the Early Universe II , Geneva, Switzerland	2026
The Growth of Galaxies in the Early Universe – XI , Sesto, Italy	2026
Seminar , Space Telescope Science Institute (StSci)	2025
Colloquium , Steward Observatory, University of Arizona	2025
The Growth of Galaxies in the Early Universe – X , Sesto, Italy	2025
Colloquium , University of St Andrews	2024
The Growth of Galaxies in the Early Universe – IX , Sesto, Italy	2024

The Growth of Galaxies in the Early Universe – VIII , Sesto, Italy	2023
The Growth of Galaxies in the Early Universe – VII , Sesto, Italy	2022
Seminar , CfA Galaxy Cluster Group, Harvard University	2022

Contributed Talks

Continuing the JWST Revolution: Understanding Early Galaxy Formation , MIAPbP, Garching, Germany	2026
Charting Cosmic Dawn in Copenhagen , Copenhagen, Denmark	2026
The Inaugural Cosmic Frontier Center Conference , UT Austin, Austin, TX, USA	2025
Beyond the Edge of the Universe , Sintra, Portugal	2024
Observing and Simulating Galaxy Evolution in the Era of JWST , Ascona, Switzerland	2024
Cosmic Dawn at High Latitudes , Stockholm, Sweden	2024
First Stars VII , New York, USA	2024
National Astronomy Meeting , Cardiff, UK	2023
First Light , MIT, Cambridge, MA, USA	2023
A new era in extragalactic astronomy: early results from the James Webb Space Telescope , University of Cambridge, Cambridge, UK	2023
National Astronomy Meeting , Warwick, UK	2022

Press

Scientific American , Astronomers puzzle over early origins of mysterious ‘red monster’ galaxy, link	2026
Physics magazine , JWST Sees More Galaxies than Expected, link	2024
Physicsworld , JWST spectrometer refines redshifts of distant galaxies, link	2023
Le Parisein , Distant galaxies: the James Webb Telescope defies all odds, link	2023
Nature News , Four revelations from the Webb telescope about distant galaxies, link	2022
BBC News , Scottish astronomers push James Webb deeper back in time, link	2022
VICE , A Cosmic Web Connecting the Universe Shapes Dark	2022

Matter in Galaxies, Study Finds, [link](#)

Teams & Memberships

Observing Collaborations

SPAM , Team member	2026 - present
Spectroscopic confirmation of the most robust $z = 15$ galaxy candidate , PI	2025 - present
CAPERS , Core-team member and survey organizer	2024 - present
CEERS , Team member	2024 - present
PRIMER , Core-team member	2021 - present
AURORA , Target selection	2021 - present

Journal clubs/seminars

EURECA galaxy seminar series organizer , Steward Observatory/NOIRLab	2025 - present
XGAL journal club organizer , University of Edinburgh	2023 - 2024

Professional Memberships

AAS Full member , American Astronomical Society	2025 - present
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Computing

Languages

Python (numpy, scipy, astropy, matplotlib)	Expert
Fortran 77/95	Proficient
C++	Experienced

Software

JWST Pipeline (NIRSpec reduction pipeline for MSA and fixed slit)	Expert
Image reduction/processing (SWarp, SCAMP, PENCIL, PSF convolution)	Expert
Photometry (SExtractor, astropy, TPHOT)	Expert
SED fitting (EAZY, BAGPIPES)	Expert

Observation Planning

Survey design of JWST/NIRSpec programs (PID: 1914, 2750, 5343, 6368, 6954)

Target Selection (by constructing catalogues from JWST/NIRCam)

Planning (MSA planning tool, exposure time calculator)

Publications (43, 6 as first author, h-index: 31, citations: 4,476)

First Author:

1. **Donnan, C.T.**, McLeod, D. J., McLure, R. J., et al., *ApJ*, 933, 224, Spectroscopic Confirmation of a Large and Luminous Galaxy with Weak Emission Lines at $z = 13.53$ 2026
1. **Donnan, C.T.**, Dickinson, M., Taylor, A. J., et al., *ApJ*, 933, 224, Very bright, very blue, and very red: JWST CAPERS analysis of highly luminous galaxies with extreme UV slopes at $z = 10$ 2025
2. **Donnan, C.T.**, Dunlop, J. S., McLure, R. J., et al., *MNRAS*, 539, 2409, No evidence (yet) for increased star-formation efficiency at $z = 6 - 13$ 2025
3. **Donnan, C.T.**, McLure, R. J., Dunlop, J. S., et al., *MNRAS*, 533, 3222, JWST PRIMER: A new multi-field determination of the evolving galaxy UV luminosity function at redshifts $z \simeq 9 - 15$ 2024
4. **Donnan, C.T.**, McLeod, D. J., McLure R. J., et al., *MNRAS*, 520, 4554, The abundance of $z \geq 10$ galaxy candidates in the HUDF using deep JWST NIRCам medium-band imaging. 2023
5. **Donnan, C.T.**, McLeod, D. J., Dunlop J. S., et al., *MNRAS*, 518, 6011, The evolution of the galaxy UV luminosity function at redshifts $z \simeq 8 - 15$ from deep JWST and ground-based near-infrared imaging. 2023
6. **Donnan, C.T.**, Tojeiro, R., Kraljic, K., *Nature Astronomy*, 6, 599, The role of the cosmic web in the scatter of the galaxy stellar mass - gas metallicity relation. 2022

Second Author:

7. McLeod, D. J., **Donnan, C.T.**, McLure R. J, et al., *MNRAS*, 527, 5004, The galaxy UV luminosity function at $z \simeq 11$ from a suite of public JWST ERS, ERO, and Cycle-1 programs 2024

Co-Author:

8. Liu, F.-Y. F., McLeod D. J., Sutcliffe B. J., et al. *arXiv:2608.07461*, A clear detection of proper motion confirms that the claimed $z \simeq 32$ galaxy candidate, “Capotauro”, is a Y-type brown dwarf 2026
8. Taylor E., Carnall A. C., Matlby, D., et al. *MNRAS*, 549, 918, The JWST EXCELS survey: outflows in $1.5 < z < 5$ quiescent and recently quenched galaxies are likely relics from episodic AGN activity 2026
12. Sanders, R. L., Shapley, A. E., et al. *ApJ*, 2003, 228, The AURORA Survey: High-Redshift Empirical Metallicity Calibrations from Electron Temperature Measurements at $z = 2 - 10$ 2026

12. Leung, H-H, Carnall A. C, Taylor, E., et al. *MNRAS*, 549, 827, 2026
[The JWST EXCELS survey: the ages and abundances of \$3 < z < 5\$ massive quiescent galaxies show that downsizing was already in place by \$z \simeq 4\$](#)
13. Barrufet, L., Dunlop, J. S., Begley, R., et al. *MNRAS*, 548, 770, 2026
[Strength in Numbers: Red Galaxies Bolster the Cosmic Star Formation Rate Density at \$z > 3\$](#)
13. Galois, P., Benoist, C., Castignani, G, et al. 2026
[arXiv:2606.13359, Galaxy clusters in the VIDEO fields: detection and characterisation in the context of MOONRISE](#)
13. Chworowsky, K., Finkelstein, S. L., Taylor, A. J., et al. 2026
[arXiv:2605.13966, Massive Galaxies Form Early and Gray: Stellar Assembly and Dust Attenuation at \$z > 3.5\$ from CAPERS](#)
13. Barro, G., Pérez-González, P. G., et al. *ApJ*, 1003, 96, 2026
[From “The Cliff” to “Virgil”: Mapping the Spectral Diversity of Little Red Dots with JWST/NIRSpec](#)
11. Napolitano, L., Pentericci, L., Dickinson, M., et al. *A&A*, 708, 102, 2026
[Ly \$\alpha\$ visibility from \$z = 4.5\$ to 11 in the UDS field: evidence for a high neutral hydrogen fraction and small ionized bubbles at \$z \sim 7\$](#)
11. Stanton, T. M., Cullen, F., Carnall, A. C., et al. *MNRAS*, 547, 449, 2026
[The JWST EXCELS Survey: gas-phase metallicity evolution at \$2 < z < 8\$](#)
11. McLeod, D. J., Dunlop, J. S., McLure, R. J., et al. 2026
[arXiv:2604.16666, A search for the first galaxies across \$0.6 \text{ deg}^2\$ of JWST imaging: new evidence for a rapid decline in star-formation activity at \$z > 12\$](#)
11. Pérez-González, P. G., Barro, G., Carniani, S., et al. 2026
[arXiv:2602.20247, Little Red Dots: One Photometric Tag Concealing Diverse Spectroscopic Flavors of Massive Star Formation and Black Hole Activity](#)
11. Berg, D. A., Sanders, R. L., Shapley, A. E., et al. *ApJ*, 996, 68, 2026
[The AURORA Survey: Robust Helium Abundances at High Redshift Reveal a Subpopulation of Helium-enhanced Galaxies in the Early Universe](#)
8. Pahl, A. J., Shapley, A., Reddy, N. A., et al. *ApJ*, 996, 37, 2025
[The JWST EXCELS Survey: A spectroscopic investigation of the ionizing properties of star-forming galaxies at \$1 < z < 8\$](#)
9. Begley, R., McLure, R. J., Cullen, F., et al. *MNRAS*, 545, 1995, 2025
[The JWST EXCELS Survey: A spectroscopic investigation of the ionizing properties of star-forming galaxies at \$1 < z < 8\$](#)
10. Stevenson, S. D., Carnall, A. C., Leung, H., et al. *MNRAS*, 545, 2087, 2025
[PRIMER & JADES reveal an abundance of massive quiescent galaxies at \$2 < z < 5\$](#)

25. Liu, F.-Y., Dunlop, J. S., McLure, R. J., et al., *MNRAS*, 545, 1961, [JWST PRIMER: A deep JWST study of all ALMA-detected galaxies in PRIMER COSMOS – dust-obscured star-formation history back to \$z \simeq 7\$](#) 2025
25. Nakane, M., Kokorev, V., Fujimoto, S., et al., *arXiv:2511.14483*, [VENUS: A Strongly Lensed Clumpy Galaxy at \$z \sim 11 - 12\$ behind the Galaxy Cluster MACS J0257.1-2325](#) 2025
30. Leung, G. C. K., Finkelstein, S. L., Pérez-González, P. G., et al., *ApJ*, 992, 26, [Exploring the Nature of Little Red Dots: Constraints on AGN and Stellar Contributions from PRIMER MIRI Imaging](#) 2024
15. Sanders, R. L., Shapley, A. E., Topping, M. W., et al. *ApJ*, 989, 209, [The AURORA Survey: The Nebular Attenuation Curve of a Galaxy at \$z = 4.41\$ from Ultraviolet to Near-infrared Wavelengths](#) 2025
14. Taylor, A. J., Kokorev, V., Kocevski, D. D., et al. *ApJL*, 989,L7, [CAPERS-LRD-z9: A Gas-enshrouded Little Red Dot Hosting a Broad-line Active Galactic Nucleus at \$z = 9.288\$](#) 2025
20. Kokorev, V., Chávez Ortiz, Óscar A., et al. *ApJL*, 988, L10, [CAPERS Observations of Two UV-bright Galaxies at \$z > 10\$. More Evidence for Bursting Star Formation in the Early Universe](#) 2025
18. Arellano-Córdova, K. Z., Cullen, F., Carnall, A. C., et al. *MNRAS*, 540, 2991, [The JWST EXCELS survey: direct estimates of C, N, and O abundances in two relatively metal-rich galaxies at \$z \simeq 5\$](#) 2025
17. Hamadouche, M. L., McLure, R. J., Carnall, A. C., et al. *MNRAS*, 541, 463, [JWST PRIMER: strong evidence for the environmental quenching of low-mass galaxies out to \$z \simeq 2\$](#) 2025
19. Cullen, F., Carnall, A. C., Scholte, D., et al. *MNRAS*, 540, 2176, [The JWST EXCELS survey: an extremely metal-poor galaxy at \$z = 8.271\$ hosting an unusual population of massive stars](#) 2025
22. Taylor, A. J., Finkelstein, S. L., Kocevski, D. D., et al. *ApJ*, 986, 165, [Broad-Line AGN at \$3.5 < z < 6\$: The Black Hole Mass Function and a Connection with Little Red Dots](#) 2025
21. Scholte, D., Cullen, F., Carnall, A. C., et al. *MNRAS*, 540, 1800, [The JWST EXCELS survey: probing strong-line diagnostics and the chemical evolution of galaxies over cosmic time using \$T_e\$ -metallicities](#) 2025
23. Kocevski, D. D., Finkelstein, A. L., Barro G., et al., *ApJ*, 986, 126, [The Rise of Faint, Red AGN at \$z > 4\$: A Sample of Little Red Dots in the JWST Extragalactic Legacy Fields](#) 2025
24. Lines, N. E. P., Bowler, R. A. A., Adams, N. J., et al., *MNRAS*, 539, 2685, [JWST PRIMER: a lack of outshining in four normal \$z = 4-6\$ galaxies from the ALMA-CRISTAL Survey](#) 2025

26. Begley, R., McLure, R. J., Cullen, F., et al. *MNRAS*, 537, 3245, The evolution of [OIII]+H β equivalent width from $z \simeq 3 - 8$: implications for the production and escape of ionizing photons during reionization 2025
27. Shapley, A. E., Sanders, R. L., Topping, M. W., et al. *ApJ*, 981, 167, The AURORA Survey: An Extraordinarily Mature, Star-forming Galaxy at $z \sim 7$ 2025
29. Shapley, A. E., Sanders, R. L., Topping, M. W., et al., *ApJ*, 980, 242, The AURORA Survey: A New Era of Emission-line Diagrams with JWST/NIRSpec 2025
28. Stanton, T. M., Cullen, F., Carnall, A. C., et al. *MNRAS*, 537, 1735, The JWST EXCELS survey: tracing the chemical enrichment pathways of high-redshift star-forming galaxies with O, Ar, and Ne abundances 2025
31. Merlin, E., Santini, P., Paris, D, et al., *arXiv:2409.00169*, ASTRODEEP-JWST: NIRCам-HST multiband photometry and redshifts for half a million sources in six extragalactic deep fields 2024
32. Carnall, A. C., Cullen, F., McLure, R. J., et al., *MNRAS*, 534, 325, The JWST EXCELS survey: Too much, too young, too fast? Ultra-massive quiescent galaxies at $3 < z < 5$ 2024
33. Stanton, T. M., Cullen, F., McLure, R. J., et al., *MNRAS*, 532, 3102, The NIRVANDELS Survey: the stellar and gas-phase mass-metallicity relations of star-forming galaxies at $z = 3.5$ 2024
34. Varadaraj, R. G., Bowler, R. A. A., Jarvis, M. J., et al., *MNRAS*, 533, 3724, The sizes of bright Lyman-break galaxies at $z \simeq 3 - 5$ with JWST PRIMER 2024
35. Weibel, A., Oesch, P. A., Barrufet, L., et al., *MNRAS*, 533, 1808, Galaxy Build-up in the first 1.5 Gyr of Cosmic History: Insights from the Stellar Mass Function at $z \sim 4 - 9$ from JWST NIRCам Observations 2024
36. Begley, R., Cullen, F., McLure, R.J., *MNRAS*, 527, 4040, Connecting the escape fraction of Lyman-alpha and Lyman-continuum photons in star-forming galaxies at $z \simeq 4 - 5$ 2024
37. Cullen, F., McLeod, D. J., McLure, R.J., *MNRAS*, 531, 997, Evidence for the emergence of dust-free stellar populations at $z > 10$ 2024
38. Arrabal Haro, P., Dickinson, M., Finkelstein, S. L., *Nature*, 622, 707, Confirmation and refutation of very luminous galaxies in the early Universe 2023
39. Carnall, A. C., McLure, R. J., Dunlop, J. S., *Nature*, 619, 716, A massive quiescent galaxy at redshift 4.658 2023
40. Hamadouche, M. L., Carnall, A. C., McLure, R. J., *MNRAS*, 521, 5400, The connection between stellar mass, age, and quenching time-scale in massive quiescent galaxies at $z \simeq 1$ 2023

41. Carnall, A. C., McLeod, D. J., McLure, R. J., *MNRAS*, 520, 3974, [A surprising abundance of massive quiescent galaxies at \$3 < z < 5\$ in the first data from JWST CEERS](#) 2023
42. Cullen, F., McLure, R. J., McLeod, D. J., *MNRAS*, 520, 14, [The ultraviolet continuum slopes \(\$\beta\$ \) of galaxies at \$z \simeq 8 - 16\$ from JWST and ground-based near-infrared imaging](#) 2023
43. Carnall, A. C., Begley, R., McLeod, D. J., *MNRAS*, 518, 45, [A first look at the SMACS0723 JWST ERO: spectroscopic redshifts, stellar masses, and star-formation histories](#) 2023